SUMMARY

- Displacement current can be defined as 'the current which comes into play in the region in which the electric field and the electric flux are changing with time'.
- Maxwell modified Ampere's law as

$$\oint \vec{B} \cdot \vec{dl} = \mu_o i = \mu_o \left(i_c + i_d \right).$$

- An electromagnetic wave is radiated by an accelerated charge which propagates through space as coupled electric and magnetic fields, oscillating perpendicular to each other and to the direction of propagation of the wave.
- Electromagnetic waves are non-mechanical and do not require any medium for propagation.
- The instantaneous magnitude of the electric and magnetic field vectors in electromagnetic wave are related by $c = E/B$.
- Electromagnetic waves are transverse in nature. This means that the oscillating electric field vector, oscillating magnetic field vector and propagation vector are (gives direction of propagation) mutually perpendicular to each other.
- Electromagnetic waves can show interference, diffraction and also can be polarized
- Electromagnetic waves carry not only energy and momentum but also angular momentum.
- Types of spectra: emission and absorption spectra.
- When the spectrum of self luminous source is taken, we get emission spectrum. Each source has its own characteristic emission spectrum. The emission spectrum can be divided into three types: continuous, line and band.
- When the spectrum obtained from the Sun is examined, it consists of a large number of dark lines (line absorption spectrum). These dark lines in the solar spectrum are known as Fraunhofer lines.

UNIT-5(XII-Physics_Vol-1).indd 280 14-02-2020 20:11:26